CNC MACHINE OPERATIONS & CARE

CNC Safety Manual

Machine hazards • operating precautions • maintenance safety

DOCUMENT CONTROL	DETAIL
Document	CNC_Safety_Manual.pdf
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Applicability	General CNC machining centres, lathes and routers; adapt locally

Important: This is an educational, generic reference. CNC machines differ substantially by builder, model, control, options, tooling and site rules. The machine-specific operator, maintenance, electrical and safety manuals take precedence. Do not use example values, procedures or alarm descriptions as substitutes for approved machine documentation.

1. Scope and authority

This manual summarizes general CNC safety principles for operators, setters, programmers and maintenance personnel. It does not replace the machine-specific risk assessment, operator manual, safety circuit design, site procedures, applicable standards or local law. The employer/site owner must provide machine-specific training and confirm safeguards are suitable.

The machine builder's instructions and site risk assessment take precedence. Never operate a machine with a known failed guard, interlock, emergency stop or other safety function.

2. Main hazards

Hazard	How injury or damage can occur	Core control
Rotating spindle / workpiece	Entanglement, impact, ejection of workpiece or tooling.	Keep doors/guards closed; secure workholding and tools; no gloves near rotating parts; wait for complete stop.
Axis and tool movement	Crushing, trapping, shearing or unexpected movement during setup/recovery.	Stay outside guarded envelope; use approved setup mode and enabling devices; never reach into moving machine.
Flying chips / broken tools	Eye/face injury, burns and cuts.	Use enclosure and specified eye/face protection; keep doors closed during cutting; remove chips with designated tools.
Sharp chips / hot parts	Cuts and burns during handling.	Allow cooling; use suitable chip hooks/tongs and gloves only when machine is stopped and risk assessment permits.
Electrical energy	Shock, arc flash, burns or fatal injury.	Keep electrical cabinets closed; only qualified persons work on electrical systems under approved isolation procedures.
Hydraulic / pneumatic energy	Injection injury, hose whip, crushing and unexpected motion.	Inspect for leaks from a safe distance; isolate and depressurize before service; never search for leaks with hands.
Coolant / mist / chemicals	Skin/eye irritation, inhalation exposure and dermatitis.	Use approved fluid, ventilation/mist control, hygiene, SDS controls and suitable PPE.
Noise / vibration	Hearing damage and fatigue.	Use enclosure, maintain tools and equipment, assess exposure and wear hearing protection where required.
Lifting / manual handling	Crush injuries, dropped loads and strains.	Use rated lifting points/equipment, trained operators and approved lifting plans.

3. General operating rules

- Operate only if trained and authorized for the specific machine and task.
- Wear fitted clothing; secure long hair and remove jewelry, lanyards and loose items that can entangle.
- Wear required safety footwear and eye protection; add hearing, face, respiratory or chemical protection as the risk assessment requires.
- Keep doors and guards closed during automatic operation. Never bypass, tape, defeat or defeat-test a safety device.
- Verify the workpiece, fixture, chuck jaws, toolholder and tool are correctly rated, installed and secured.
- Use the correct program, work offset, tool data, units and coordinate system. Verify new or edited programs using the approved prove-out method.
- Keep the enclosure and floor clear. Do not leave keys, tools, gauges or loose objects where they can become projectiles.

- Never leave an operating machine unattended unless the machine and site procedure explicitly allow it.
- Do not use compressed air to clean clothing or skin. Control chip removal and coolant spills using approved methods.

4. Setup, proving and first-piece checks

1. Review the job risk assessment, drawing, setup sheet, tool list and program revision.
2. Inspect the machine, workholding, tooling and guarding. Confirm no damage or obstruction is present.
3. Secure the workpiece and verify chuck/fixture engagement, clamping pressure and tool clearance per OEM instructions.
4. Confirm work offsets, tool length/radius data, units, plane, spindle direction and coolant requirements.
5. Use the machine's approved graphics, simulation, single-block, reduced-rapid or dry-run features as specified by the procedure.
6. Keep all personnel outside the guarded envelope and maintain a clear view of the machine during prove-out.
7. Inspect the first part using the approved measurement plan before releasing production.

5. Safety during maintenance and cleaning

> Maintenance, jam clearing, tool-change recovery, chip conveyor clearing and access behind guards can expose personnel to hazardous energy. Apply the formal site LOTO procedure; the following is a planning aid, not a complete isolation instruction.

1. Identify every energy source: electrical, pneumatic, hydraulic, gravity, spring, thermal, spindle rotation and stored mechanical energy.
2. Notify affected persons and shut down using the prescribed sequence.
3. Isolate each energy source with approved devices and apply personal locks/tags under site rules.
4. Dissipate or restrain stored energy (e.g., pressure, suspended components, rotating parts) using the OEM procedure.
5. Verify isolation and zero-energy state with the approved test method before work begins.
6. Perform the work using suitable tools, PPE and lifting aids; maintain control of removed guards and fasteners.
7. Before restoring energy, inspect the area, reinstall guards, remove tools/materials, account for personnel and notify affected persons.
8. Remove locks/tags only by the person and process authorized by site LOTO rules; perform required safety and functional tests.

6. Emergency response

- Know the location and operation of emergency stops, fire alarms, first-aid supplies, eyewash and emergency exits.
- For imminent danger, activate the appropriate emergency stop and site emergency response. An E-stop is not an energy-isolation device.
- Do not open a guard or enter the machine until all motion has stopped and hazards are controlled.
- For fire, electrical incident, chemical exposure or serious injury, follow the site emergency plan and call trained responders.
- Do not restart after an emergency, collision or safety trip until the cause is assessed and an authorized person releases the machine.

7. PPE and personal hygiene

- Minimum PPE is determined by the site risk assessment and local regulations; common requirements include safety footwear and eye protection.
- Use hearing protection where exposure assessment requires it.
- Select chemical-resistant gloves for coolant/cleaning chemicals according to the SDS; do not wear gloves near rotating machinery.
- Wash exposed skin after coolant contact; remove contaminated clothing and report persistent irritation.
- Use respiratory protection only under a proper respiratory protection program, with suitable selection, fit and training.

8. Prohibited practices

- Operating with a guard removed, interlock defeated or safety device malfunctioning.
- Reaching into the machine while the spindle or axes can move.
- Measuring, deburring or adjusting a workpiece while it is rotating or while hazardous motion is possible.
- Clearing chips or jams without the required stop, isolation and approved tools.
- Using damaged, unbalanced, incorrectly rated or improperly secured tools/workholding.
- Changing safety, servo, spindle or PLC parameters without authorization.
- Using a phone or becoming distracted during setup, proving or operation.

9. Pre-start operator checklist

Check	Pass / fail / N/A
Operator trained and authorized; correct job documentation available.	_____
Machine condition acceptable; no leaks, damage or abnormal noise.	_____
Guards, doors, interlocks and E-stops pass the approved check.	_____
Workpiece, fixture, chuck and tools correctly secured and rated.	_____
Correct program revision, units, offsets and tool data verified.	_____
Coolant, lubrication, air and other required utilities in range.	_____
Enclosure clear; chip bin/conveyor ready; floor safe and unobstructed.	_____
Required PPE worn; emergency arrangements understood.	_____

10. Training and authorization record

Person	Machine / task	Training date	Assessor / authorization
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

11. Reference and review

The site should maintain current copies of: (1) machine builder’s operator and maintenance manuals; (2) control-system alarm manual; (3) electrical, pneumatic and hydraulic schematics; (4) risk assessment and LOTO procedure; (5) coolant and chemical SDS; (6) lifting and workholding instructions; and (7) applicable local occupational safety requirements. Review this document after a machine change, incident, new process, revised regulation or periodic safety review.